

PERFORMANCE ENHANCEMENT OF SOLAR AIR HEATERS USING PASSIVE FLOW CONTROLLERS: EXPERIMENTAL STUDY WITH LOUVERED SHEETS

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ABSTRACT

Passive flow control devices have been extensively utilized in solar air heaters to enhance convective heat transfer by promoting turbulence and fluid mixing. Conventional designs such as vortex generators, ribs, and winglets have demonstrated notable improvements in thermal performance; however, these often come with increased pressure losses. To explore alternative flow enhancement strategies, the present study introduces louvered sheets as a novel passive flow control element for solar air heaters.

Two louvered-sheet configurations were experimentally investigated. In the first configuration, two louvered sheets were positioned at one-third (1/3) and two-thirds (2/3) of the test section height, while in the second configuration, they were installed at one-quarter (1/4) and three-quarters (3/4) of the test section height. Experiments were conducted using a custom-built test facility at the Guimarães Instrumentation, Measurements, and Aerodynamic Sensing (GIMAS) Laboratory at The Pennsylvania State University.

The thermal-hydraulic performance of both configurations was assessed using the Thermal Hydraulic Performance Factor (THPF), which accounts for both Nusselt number enhancement and pressure losses through the friction factor. The results indicate that the second configuration provides superior performance, achieving a maximum THPF of approximately 1.74 at the lowest Reynolds number around 3000, demonstrating its greater efficiency in balancing heat transfer improvement and flow resistance.

KEYWORDS: (Thermal Hydraulic Performance, Vortex Generators, Solar Air Heaters, heat transfer enhancement)

1. INTRODUCTION

Global energy demands are rising constantly, which is increasing our dependency on fossil fuels and creating drastic climatic changes due to which there has been a worldwide effect to adopt more renewable sources of energy. Among all different renewable sources, solar energy tends to stand out the most due to its abundance, versatility and scalability for both regular and industrial purposes. However, there tend to be a problem with these solar thermal systems, they have low heat transfer efficiency due to the inherently poor thermal conductivity and specific heat of air [1]. Therefore, researchers have tried to focus on inducing turbulent flow and vortex motion absorbing channels to enhance convective thermal boundaries. Passive flow-enhancement techniques such as ribs, baffles, fins, grooves, and vortex generators (VGs) have proven to be highly effective, which offers high thermal performance without increasing the overall power consumption [2]. These devices are simple to create and maintain, which makes it perfect for large-scale implementation. Traditional vortex generators face a huge challenge due to friction loss and geometric constraints, which prompts the emergence of hybrid passive systems such as louvered sheet integrated VGs; it combines multidirectional flow redirections with a secondary vortex generator to achieve high heat coefficient [3]. Recent advancements in the field of

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additive manufacturing and topology optimized designs such as hybrid geometries help us to throw a positive light towards sustainable, high efficiency solar air-heating and heat-exchange systems [4].

The most common technique for systems utilizing solar energy heat with air as the working fluid involves inducing vortex turbulent flow. This specific approach enhances mixing between the flow and heated surface, effectively disturbing the thermal boundary layer formation and increasing the flow mixing. This mixing can be accomplished through the use of either active or passive devices which is later considered more efficient as it does not consume power. Various passive flow control methods, including fins, ribs, grooves, baffles, and vortex generators (VGs), have been subjects of studies and research. The application of these advancements extends to enhancing electronic cooling, efficiency of air heating systems like space heating and drying processes [5]. These passive flow control methods have additional advantages, such as ease of manufacturing, installation and maintenance. These are examples of passive systems that consume almost no energy when used properly.

Most common types of vortex generators rely on baffle design, whose main purpose is to enhance thermal performance by disrupting the boundaries. Baffle designs have consistently demonstrated their effectiveness in improving thermal performance. Various shapes, configurations and arrangements of baffles have been studied extensively through both computational and experimental methods. The thermal-hydraulic performance factor (THPF) is most commonly used to evaluate the effectiveness of such passive flow enhancement techniques. It is basically the ratio of the Nusselt number to the friction factor, both are normalized against the corresponding values for a smooth channel [5].

A combined computational and experimental study investigating transverse, inverse transverse, dimple, inclined dimple, arc, and sine-wave baffle vortex generators identified sine-wave baffles delivering the best performance among these designs (THPF of 2.05) [6]. However, the highest experimentally achieved THPF is around 3.9 using 60° angled broken multiple V-type baffles at a low Reynolds number of 3000 [7]. This demonstrates how modifications such as adjusting the angle or type of baffles can significantly impact the thermal hydraulic performance, highlighting the utmost importance of an optimized design [6].

The highest thermal performance of a heat transfer system depends on more than just the shape of the vortex generator (VG), these complex results are influenced significantly by factors such as the arrangement of VG arrays, the blockage ratio, and especially the placement of baffles. Baffles are most effective at boosting heat transfer when they are positioned near the inlet, specifically in the first half of the test section, where they have a critical impact on the system's overall thermal enhancement factor. Based on various experimental data we can conclude that the highest thermal performance of around 1.83 was achieved by using a single baffle-vortex generator array combined with a blockage ratio of 0.5 [8].

Delta winglet vortex generators is a common passive technique in solar air heaters used to disrupt the airflow boundary layer and create secondary vortex flows to enhance heat transfer. While the standard delta winglet VGs performed reasonably well, achieving a maximum Thermal Hydraulic Performance Factor (THPF) of 2.5 at a 30° angle of attack [9, 10, 11]. After certain experimental improvements done to the winglet shapes led to better results, especially wavy delta winglets demonstrated good performance and perforated curved delta achieved the best record for getting the maximum THPF of about 2.26 [12].

Another VG type is the longitudinal vortex generators, which are a widely studied passive flow control method used to enhance heat transfer by inducing vortex flows that disrupt the thermal boundary layer. Both computational and experimental studies have explored their performance, with optimal results observed when configured to point upward at a 60-degree angle of attack with a blockage ratio of 0.4 [13]. However, the maximum THPF obtained was only 1.17, which is relatively low compared to baffles or delta winglets. Despite this limitation, numerous studies have sought to enhance the performance of longitudinal vortex generators. For instance, incorporating a perforated concave shape has been shown to reduce the pressure drop, combined with a slight reduction in heat transfer improvement compared to non-perforated designs [14].

Despite significant advancements in passive thermal-enhancement techniques, a major limitation of many existing heat-exchange configurations is that they manipulate the flow primarily in a single direction. A more effective recent approach involves integrating louvered sheets with vortex-generating features, thereby creating a hybrid passive system that combines two complementary enhancement mechanisms—cross-flow induction from louvers and vortex generation—to improve heat transfer without requiring any external power input.

Louvered sheets consist of thin metallic or polymeric fins with slanted or angled openings that redirect air

between adjacent channels, promoting cross-flow mixing and disrupting the thermal boundary layer. While previous studies have demonstrated that combining louvered sheets with longitudinal or delta-winglet vortex generators can induce strong secondary vortices and enhance local mixing, the present work isolates the effect of louvered sheets alone, without incorporating additional vortex generators in the experimental setup.

This configuration increases the effective heat-transfer rate by re-energizing the flow downstream, sustaining higher turbulence levels, and improving fluid–surface contact. As a result, the system achieves elevated Nusselt numbers with only minor pressure penalties, illustrating the potential of louvered-sheet architectures as an efficient passive enhancement technique.

Promvonge et al. [15] showed that winglet-type vortex generators generate strong longitudinal vortices that intensify near-wall mixing and significantly increase local Nusselt numbers. When louver structures are integrated with winglet surfaces, as in our cooperating group's study [3], the combined geometry produces both crossflow redirection and sustained longitudinal vortex pairs, yielding higher thermo-hydraulic performance than conventional delta-winglets. Further [7] reported that radially arranged winglet fins around tubes improve heat-transfer performance by strengthening secondary vortex structures, supporting the effectiveness of hybrid louver–winglet designs. Study [3] has demonstrated that louvered-winglet vortex generators can improve THPF by 20–30% over traditional delta-winglets with the same existing geometries suitable for solar air heaters.

Louvered fins have gained a lot of attention nowadays as it helps us to significantly improve the thermal efficiency of solar air heaters. These fins are designed such a way that it helps us to achieve multidirectional air redirections and cross-flow mixing between adjacent channels thereby disrupting the thermal boundary layer effectively. Various experiment has been conducted in this field where solar air heaters equipped with louvered fins spaced around 20 mm apart showed 106.7% improvement in thermal efficiency compared to a normal smooth channel design [16]. The major reason for this improvement is due to the re-energization of air flow and improved surface contact between fluid and absorber plate. Louvered fin structure is a simple design which makes it a bit easier to create and integrate into large scale heating systems with minimal cost additions.

Throughout the scope of this research every surface geometry has been made using steel and aluminum but as the world is moving towards more advanced versions of additive manufacturing technology, it can be implemented in making more complex designs and geometries. Traditional manufacturing methods like extrusion, stamping or sheet-metal forming impose various geometric constraints and it is not always possible to make every single design. On the other hand, 3-D printing enables creation of highly intricate internal flow passages within one single continuous part which enhances overall stability of the design [4]. Through various studies conducted in the past it has been proved that topology optimization along with 3-D printing can significantly improve both thermal and hydraulic performance. Past researches show that these 3D printed heat exchangers with embedded vortex generators can achieve around 32% higher Nusselt number compared to traditional designs with similar pressure conditions [4].

Advancements in additive manufacturing has definitely opened a new scope of development of high performance for passive flow enhancements in solar air heaters and heat exchange systems. Through some experiments it was proved that 3-D printed sheets (TPMS heat exchanger cores) achieved 48–61% higher heat transfer compared to conventional geometric designs due to their high surface area to volume ratios and controlled internal flow paths [17]. It was demonstrated that the dual-fluid heat exchangers through multi-layer topology optimization, could be designed to reduce pumping power while enhancing thermal exchange [18]. Multi-layer topology optimization is simply a method used to determine the best possible material distribution and internal geometry of a structure that consists of multiple interacting layers or flow domains. If we combine all the knowledge that we have in additive manufacturing, machine learning design tools and topology optimizations, it will help us to create complex internal geometric designs to improvise thermal effectiveness [19].

On the basis of various findings from past research, it is quite evident that various passive flow control strategies such as baffles, ribs and vortex generators have demonstrated notable improvements in the field of heat transfer, but still there is a lack of experimental optimization to balance enhanced Nusselt number with acceptable pressure drops. To address this major research gap and validate computational findings, an experimental setup has been established at Guimarães Instrumentation, Measurements, and Aerodynamic Sensing (GIMAS) Laboratory at The Pennsylvania State University. This facility helps to conduct systematic testing of various vortex generator configurations, louvered sheet geometries and hybrid combinations under controlled thermal and aerodynamic conditions. The main objective of this research is to identify the perfect possible design and

positioning of vortex generators that maximize heat-transfer performance while minimizing flow resistance. By implementing advanced measurement techniques and exploring new aspects of passive flow control systems, this study aims to strengthen the foundation for the future development of high-efficiency solar air heaters and compact heat exchangers capable of meeting the constantly growing renewable energy demands.

2. EXPERIMENTAL DESIGN

The experimental setup utilized in this study is illustrated in Figure 1. The air is supplied by a 2 hp electric blower (1) to the settling chamber (2), which is used to precondition the air to achieve a uniform flow. The outlet of the settling chamber is equipped with a honeycomb (3) to straighten the flow, ensuring that the air stream is smooth and well-aligned as it exits the chamber. After passing through the honeycomb, the air travels through a sufficiently long section (4) to allow the flow to fully develop before entering the test section (5). The test section is designed to be modular, allowing easy modifications to validate different geometries experimentally. Additionally, the setup provides optical access for future flow visualization experiments using particle image velocimetry, as both the upper and lower surfaces of the test section are made of glass, while the sides are constructed from aluminum. After passing through the entrance section (6), the air is released into the environment. This design ensures a controlled and adaptable environment for studying heat transfer enhancements and flow behavior in solar air-heating channels. The experimental setup consists of three main sections; entrance section, test section, and exit section that have lengths of 35", 36", and 18", respectively. The channel height is 2" and the width is 20" throughout the sections.

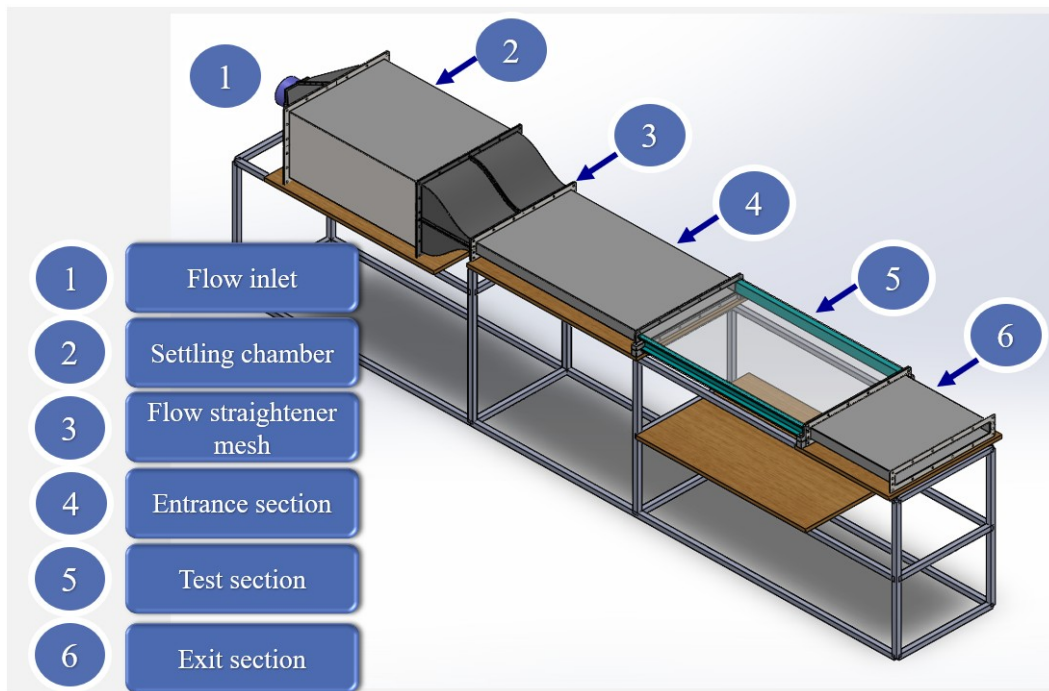


Fig. 1 Experimental rig design.

The airflow is controlled using a voltage controller connected to the 2hp air blower. It then undergoes preconditioning in a settling chamber before entering the channel, which featured a contraction ratio of 1:6.5. To ensure flow uniformity, a honeycomb flow straightener was positioned at the outlet of the settling chamber. The entrance and exit sections of the chamber, measuring 28 inches in length and 13 inches in height, are fabricated using a Qidi X-CF Pro 3D printer with carbon fiber filament and assembled using 3M structural sealant. Constructed from A36 steel gauge 12 cuts with a Langmuir Crossfire Pro CNC torch and an Everlast 62i system, the entrance and exit sections adhere to ASHRAE recommendations outlined in standard 93. The test section consists of clear tempered glass with a thickness of 15/64" and aluminum sidewalls measuring 0.89" thick. Electric heating elements are mounted on a 0.019" thick aluminum sheet, which serves as the heating source and is placed on the top surface of the test section. Both the top and bottom surfaces are modular,

allowing for easy replacement when testing different VG designs. To ensure precise temperature measurements while minimizing ambient heat losses, the test section is encased in insulation. A 1" thick ceramic fiber and wooden box provide structural support, with insulation strategically placed between the test section walls and plywood boards. The upper portion of the insulation is detachable, enabling easy access for instrumentation and adjustments, as shown in Figure 2.

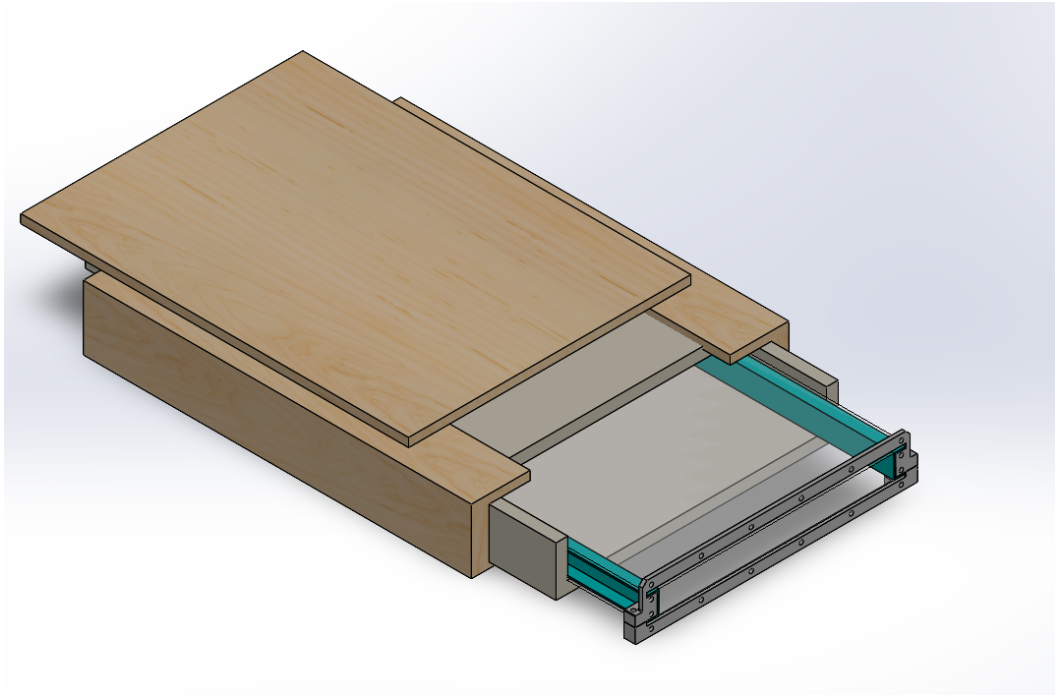


Fig. 2 Test section with the insulation and the wooden box.

To provide heat to the flow and simulate conditions similar to real solar air heaters, six rows of electric heaters were used. Each row contained five electric heaters, each with a power output of 50 W. These heaters were connected to six voltage transformers and six proportional integral derivative (PID) controllers, as shown in Figure 3 and Figure 4.

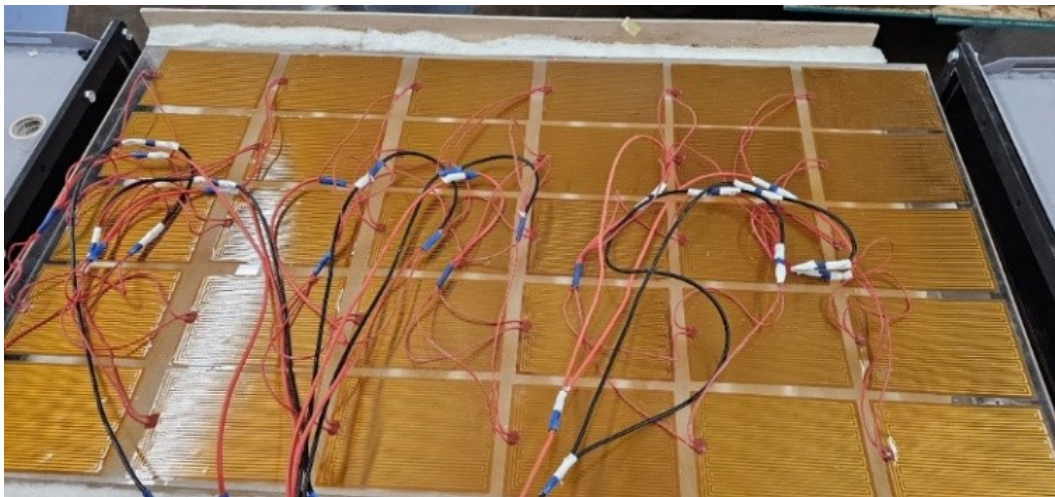


Fig. 3 Electric heating element connection on the surface of the test section.

2.1 Louvered Sheets Design Louvered sheets were tested in the channel to determine the thermal hydraulic performance of solar air heaters with the present of these louvered sheets in two configurations. The first one was placing two louvered sheets at $1/3$ and $2/3$ of the test section height. The second configuration is using two louvered sheets at $1/4$ and $3/4$ of the test section height. The louvered sheet has a design as shown in Figure 5.



Fig. 4 PID controllers, and voltage transformers.

3. DATA COLLECTION

An array of 15 thermocouples (TC) type K was positioned throughout the test section to measure local temperatures, as shown in Figure ???. In addition, four type K thermocouples were placed at the inlet and another four at the exit section to measure the inlet and exit flow temperatures. The thermocouples were connected to a monitor to read the temperature data directly.

Static pressure measurements were taken at both the inlet and outlet of the test section using two static pressure taps at each location. These readings were used to determine the pressure drop along the test section.

To evaluate the flow velocity entering the test section, dynamic pressure measurements were obtained by averaging two readings from different setups. The first setup utilized a Kiel probe positioned at the center of the test section in conjunction with two static pressure taps to determine the dynamic pressure. The second setup employed a Pitot-static probe located at a streamwise position of $x/L=0.25$

All pressure data were acquired using a high-accuracy pressure sensor (DLHR-L01D-E1BD-C-N2V8), featuring a measurement range of ± 1 inH₂O (± 250 Pa) and an accuracy of $\pm 0.25\%$ FS (± 0.62 Pa). This high precision enabled reliable data collection at low Reynolds numbers. The sensors were connected to a Raspberry Pi for data acquisition, ensuring synchronized and high-resolution sampling across all channels.

The experiments were conducted over multiple trials, with each trial consisting of 600 readings collected at an acquisition frequency of 20 Hz (one reading every 0.05 seconds). This sampling rate provided sufficient temporal resolution to capture steady and repeatable flow behavior within the test section.

For each measurement point, the mean value was calculated across all repeated trials to represent the average behavior, while the standard deviation of these measurements was computed to quantify the variability. The resulting standard deviations were used to generate error bars, which represent the measurement uncertainty in the reported results.

4. DATA REDUCTION AND ANALYSIS

To experimentally validate the collected data, an initial step involves characterizing the empty test section without any heat transfer-enhancing structures. This serves as the "base" case for subsequent comparisons with configurations featuring various geometries of flow control devices. The experimental outcomes will be presented in terms of Nusselt number (Nu) and friction factor (f). The average convection heat transfer coefficient (h) is derived from temperature measurements along the test section and the prescribed heat flux on the absorbing plate. The calculation of the convection heat transfer coefficient follows Equation 1:

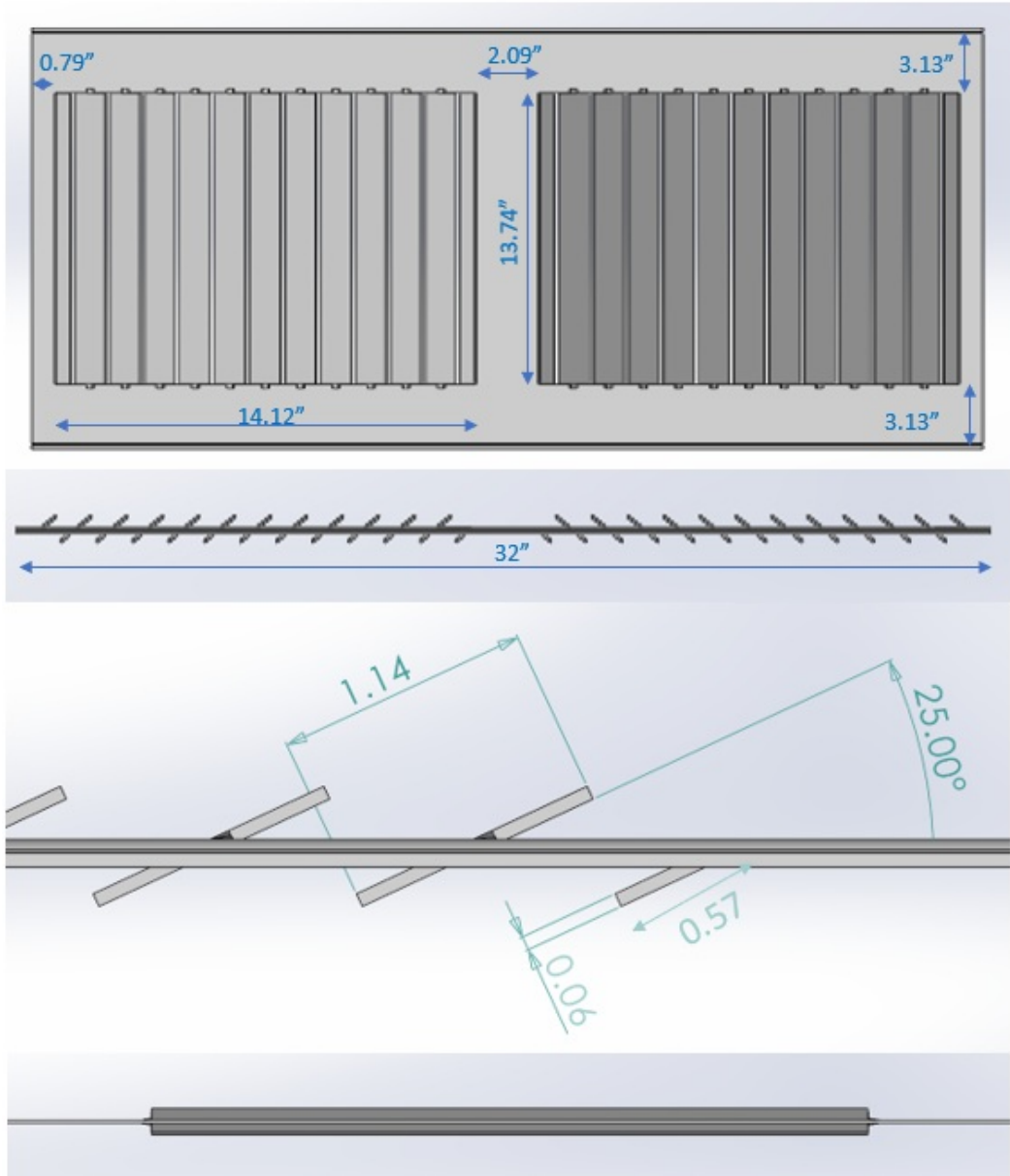


Fig. 5 Louvered sheet design.

$$h = \frac{Q_{\text{conv}}}{A(\tilde{T}_s - T_f)} \quad (1)$$

where A is the convection heat transfer area on the heated surface, and $\tilde{T}_s$ is the average temperature of the heated surface, obtained from 15 thermocouples located along the test section:

$$\tilde{T}_s = \frac{T_{s1} + T_{s2} + \dots + T_{s15}}{15} \quad (2)$$

The bulk (average) fluid temperature is defined as

$$T_f = \frac{\bar{T}_{\text{out}} + \bar{T}_{\text{in}}}{2} \quad (3)$$

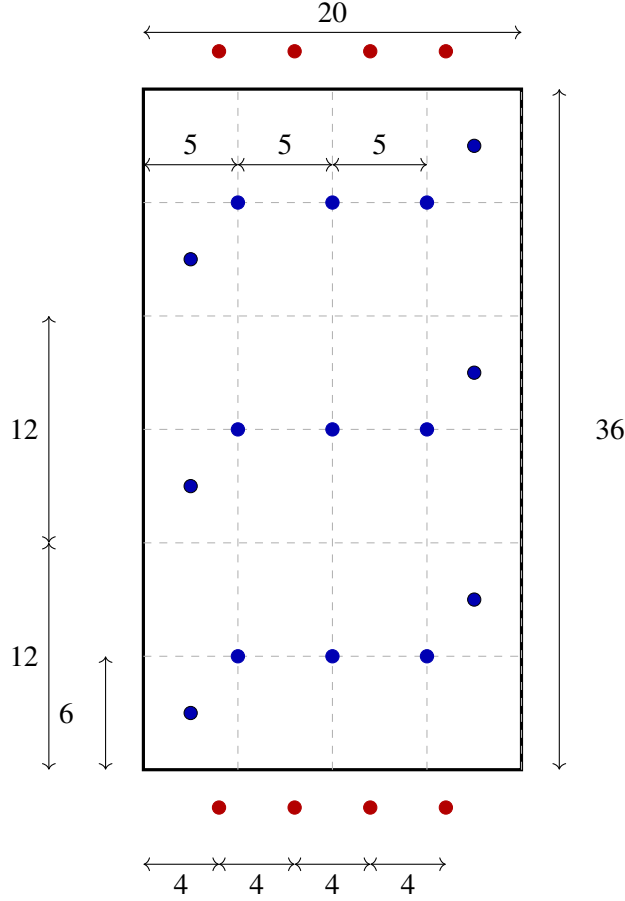


Fig. 6 Thermocouple layout on the test section (all dimensions in inches).

where $\bar{T}_{in}$ and $\bar{T}_{out}$ are the average inlet and outlet air temperatures, respectively.

The heat absorbed by the air in the test section is given by

$$Q_{conv} = Q_{air} = \dot{m}C_p(\bar{T}_{out} - \bar{T}_{in}) \quad (4)$$

The convective heat transfer is also equal to the electrical heat input minus the heat loss:

$$Q_{conv} = IV - Q_{loss} \quad (5)$$

The average Nusselt number and the friction factor in the channel are calculated as

$$Nu = \frac{hD_h}{k} \quad (6)$$

$$f = \frac{\Delta P}{\left(\frac{\rho_{air} V^2}{2}\right)} \left(\frac{D_h}{L}\right) \quad (7)$$

where ΔP is the static pressure drop between the inlet and outlet, V is the average air velocity at the inlet, L is the test section length, and D_h is the hydraulic diameter, defined as

$$D_h = \frac{2(W \times H)}{(W + H)} \quad (8)$$

To evaluate the overall performance, the thermal–hydraulic performance factor (THPF) is introduced as

$$\text{THPF} = \frac{Nu_R}{(f_R)^{1/3}} \quad (9)$$

where Nu_R and f_R are the Nusselt number ratio and friction factor ratio of the modified channel relative to the smooth channel:

$$Nu_R = \frac{Nu}{Nu_0} \quad (10)$$

$$f_R = \frac{f}{f_0} \quad (11)$$

5. UNCERTAINTY ANALYSIS

The data collected in this experiment is subject to a certain degree of uncertainty due to the inherent inaccuracies associated with the instrumentation used. Table 1 provides detailed information on the instruments employed in this study, along with their respective accuracy specifications.

Table 1 Instrument accuracy specifications.

Instrument	Accuracy
K-type thermocouples	$\pm 1.1^\circ\text{C}$
Pressure sensor (DLHR-L01D-E1BD-C-N2V8)	$\pm 0.62\text{ Pa}$
Measuring tape (width and length)	$\pm 1\text{ mm}$
Caliper (height measurement)	$\pm 0.1\text{ mm}$

To determine the uncertainties associated with the calculated parameters, the following propagation-of-error equations were used [20]:

$$R = X_1^a X_2^b X_3^c \dots X_m^m \quad (12)$$

$$\frac{\Delta R}{R} = \left[\left(\frac{a \Delta X_1}{X_1} \right)^2 + \left(\frac{b \Delta X_2}{X_2} \right)^2 + \left(\frac{c \Delta X_3}{X_3} \right)^2 + \dots + \left(\frac{m \Delta X_M}{X_M} \right)^2 \right]^{1/2} \quad (13)$$

These relationships were applied to estimate the uncertainties in the thermal and hydraulic parameters, with the maximum uncertainties summarized in Table 2.

Table 2 Maximum uncertainties of the calculated parameters.

Parameter	Maximum Uncertainty
Friction factor (f)	4.0%
Nusselt number (Nu)	6.2%
Reynolds number (Re)	2.0%
Thermal hydraulic performance factor (THPF)	6.4%

This uncertainty analysis validates the reliability of the experimental data, instilling confidence in the derived thermal and hydraulic performance metrics, which fall within the range reported in existing experimental literature [21], [22].

6. RESULTS AND DISCUSSION

6.1 Smooth-Channel Validation. The experiment began with a verification of the baseline conditions by evaluating heat transfer in the unobstructed (smooth) channel. These measurements were then compared with theoretical expectations derived from well-established correlations. In particular, the Dittus–Boelter correlation [23] was applied to estimate the Nusselt number, as presented in the following correlation.

$$\text{Nu}_{\text{Dittus-Boelter}} = 0.023 \text{Re}^{0.8} \text{Pr}^{0.4} \quad (14)$$

while the Blasius correlation [24], was applied to determine the friction factor:

$$f_{\text{Blasius}} = 0.316 \text{Re}^{-0.25} \quad (15)$$

This validation step was essential to verify that the experimental rig functioned properly and that the baseline measurements aligned with established reference data. Similar validation procedures for both the smooth channel and a configuration incorporating simple rectangular vortex generators were carried out in earlier work [5]. In the present experiment, additional improvements were implemented to enhance measurement reliability, specifically, an upgraded pressure sensor and refined pressure-measurement techniques were introduced. These enhancements provided higher accuracy and greater stability in the recorded data, thereby establishing a robust foundation for subsequent investigations involving various geometries and flow-control mechanisms.

The thermal performance of the smooth channel was evaluated by comparing the experimentally determined Nusselt numbers with theoretical predictions across a wide range of Reynolds numbers. As illustrated in Figure 7, the measured values closely followed the Dittus–Boelter correlation, exhibiting only minor deviations. These small differences are attributed primarily to uncertainties in temperature measurement, particularly at lower temperature gradients.

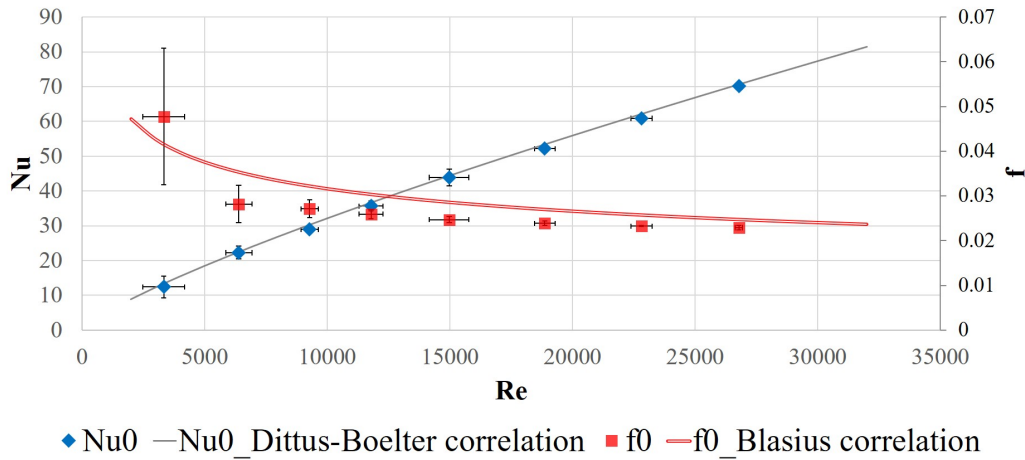


Fig. 7 Thermo-hydraulic validation of the smooth channel.

The hydraulic performance was assessed through the friction factor, which was benchmarked against the Blasius correlation for smooth ducts. The experimental results showed strong agreement with theoretical expectations, confirming that the pressure-drop behavior of the system was consistent with that of fully developed smooth-channel flow. A noticeable deviation occurred at the lowest Reynolds number, likely resulting from increased uncertainty in flow measurements at low velocities.

Overall, the close alignment of both thermal and hydraulic performance with established correlations validates the accuracy and stability of the experimental rig. This successful verification step, supported by the smooth-channel reference data shown in Figure 7, ensures confidence in the system's ability to generate reliable results in subsequent studies involving vortex generators and other passive flow-control configurations.

6.2 Thermal–Hydraulic Performance with Louvered Sheets. After completing the validation for the smooth channel, the study proceeded to investigate the thermal–hydraulic performance of the solar air heaters with the addition of louvered sheets. Two configurations were examined to evaluate their effects on heat transfer enhancement and pressure-drop characteristics. In the first configuration, two louvered sheets were installed at positions corresponding to one-third and two-thirds of the test section height. In the second configuration, the sheets were placed at one-quarter and three-quarters of the test section height, allowing for a comparative assessment of louver positioning on system performance.

The thermal performance of the first configuration is illustrated in Figure 8, which shows that integrating louvered sheets significantly increases the Nusselt number relative to the smooth-channel baseline.

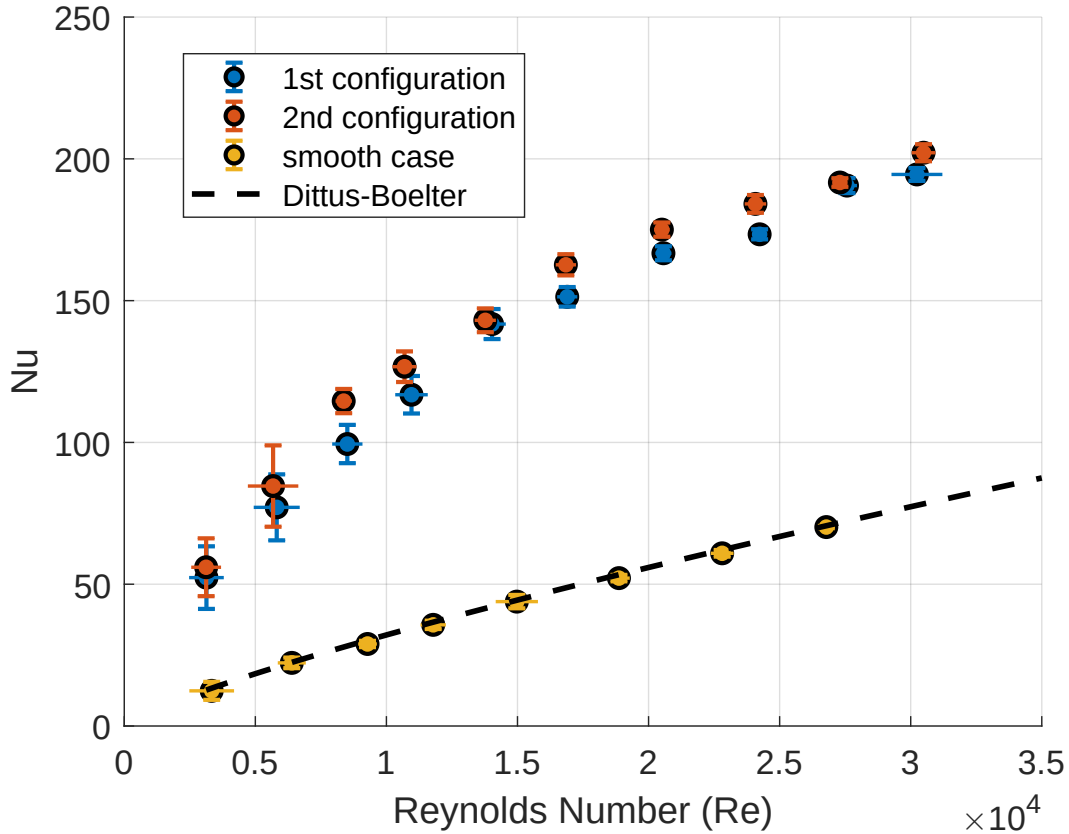


Fig. 8 Thermal performance of the two tested configurations compared to the smooth reference case.

Figure 9 presents the Nusselt number ratio (Nu_R), defined as the ratio of the modified Nusselt number to that of the smooth channel. The results show that the highest enhancement occurs at the lowest Reynolds number, with the magnitude of improvement gradually decreasing as the Reynolds number increases.

For the pressure-loss measurements, Figure 10 presents the friction-factor results for the first configuration. As expected, the introduction of louvered sheets increases the flow resistance noticeably, especially when compared with the smooth channel, which represents an almost resistance-free baseline. Consequently, the friction factor for the smooth channel remains close to zero, whereas the addition of the louvered sheets produces a significant rise in friction factor due to the increased flow obstruction and surface interaction.

The friction factor ratio (F_R), shown in Figure 11, represents the ratio of the modified friction factor relative to that of the smooth channel. The results indicate a clear trend in which F_R increases progressively with increasing Reynolds number.

The overall performance, combining both the thermal and hydraulic behaviors of the system, can be evaluated using the Thermal–Hydraulic Performance Factor ($THPF$). Figure 12 shows that the $THPF$ reaches its peak value of approximately 1.53 at the lowest Reynolds number of around 2800. As the Reynolds number increases, the $THPF$ gradually decreases, eventually falling below 1 at a Reynolds number of about 21,000. Beyond this point, the use of louvered sheets becomes thermally and hydraulically unfavorable, as the associated pressure penalties outweigh the heat-transfer benefits.

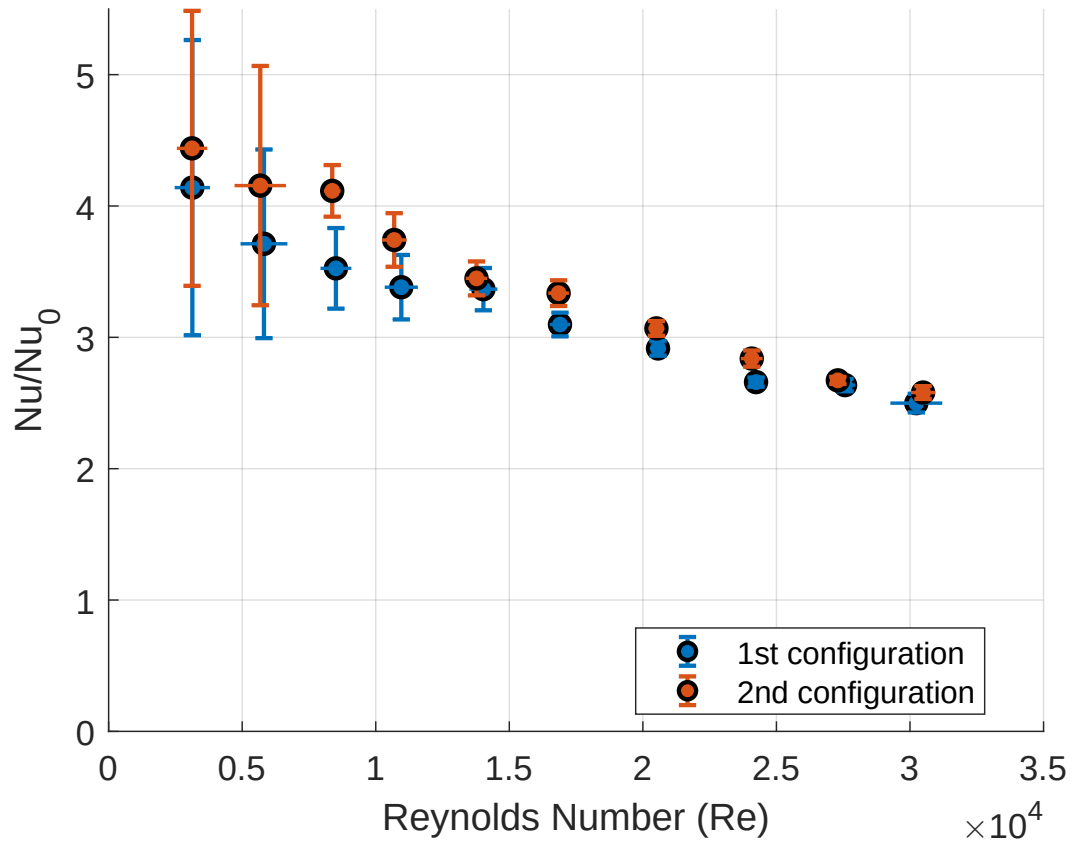


Fig. 9 Nusselt number ratio (Nu_R) for the two configurations.

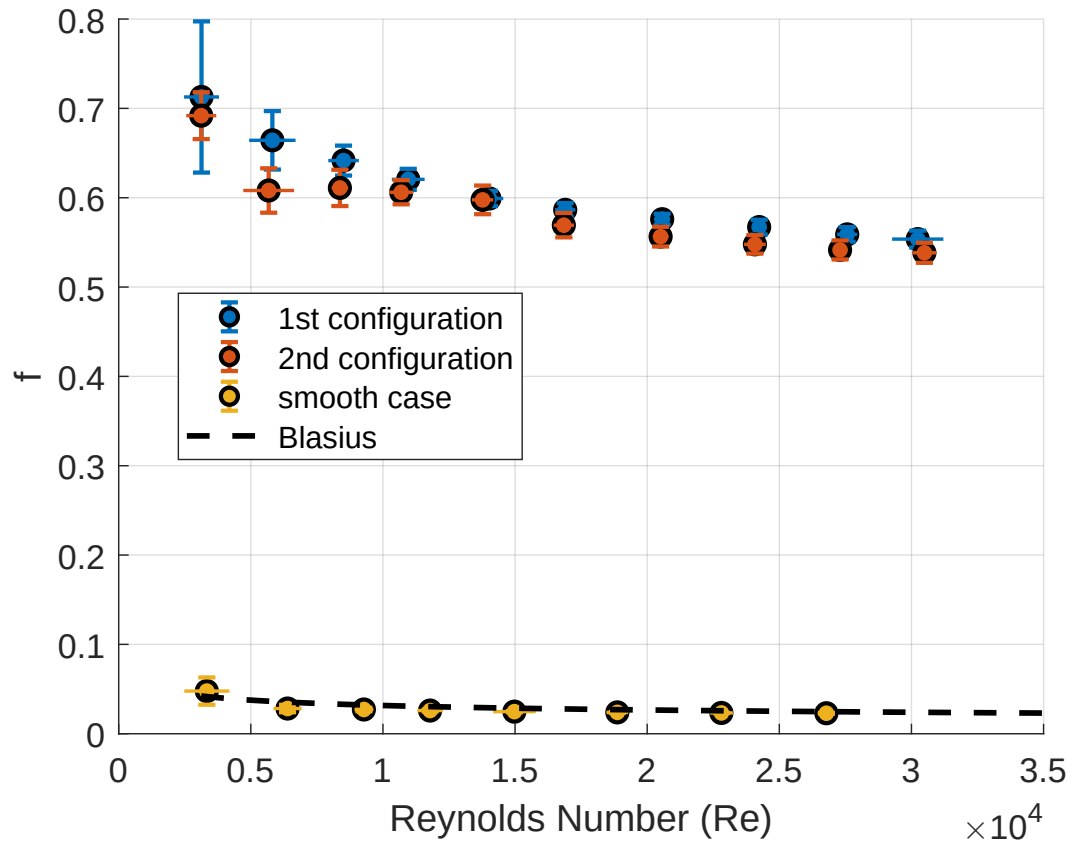


Fig. 10 Friction factor of the two tested configurations compared to the smooth reference case.

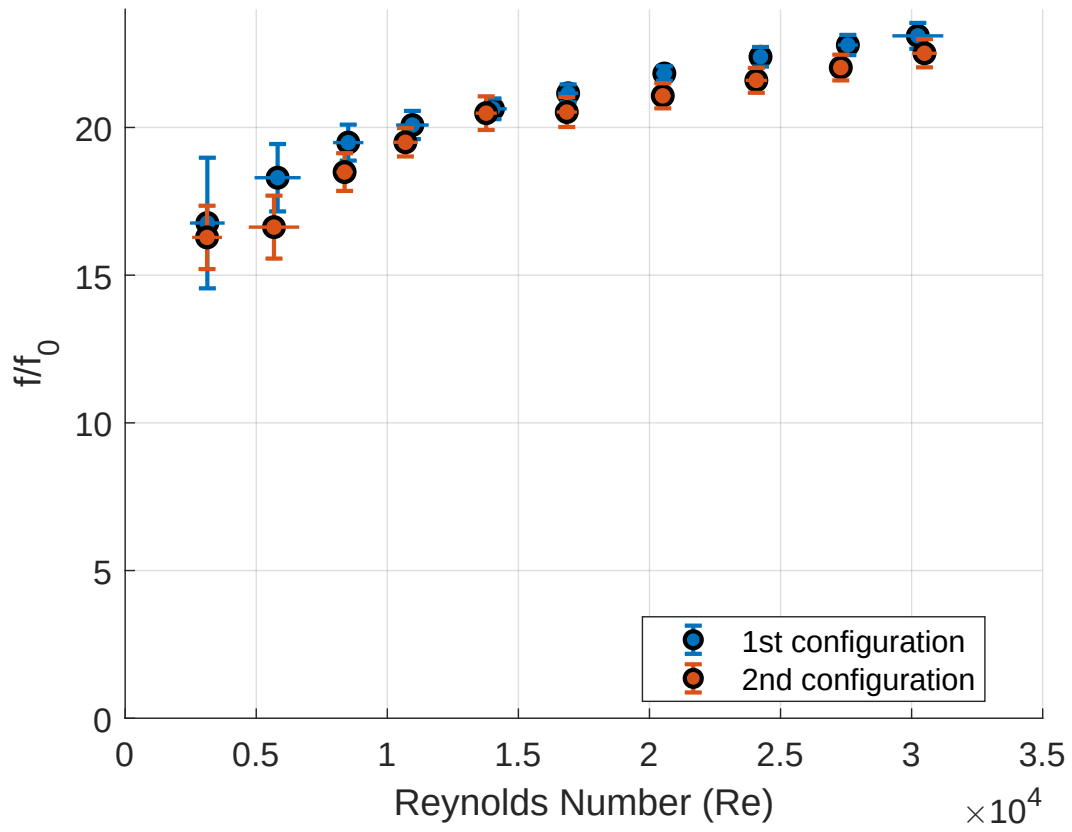


Fig. 11 Friction factor ratio (F_R) for the two configurations.

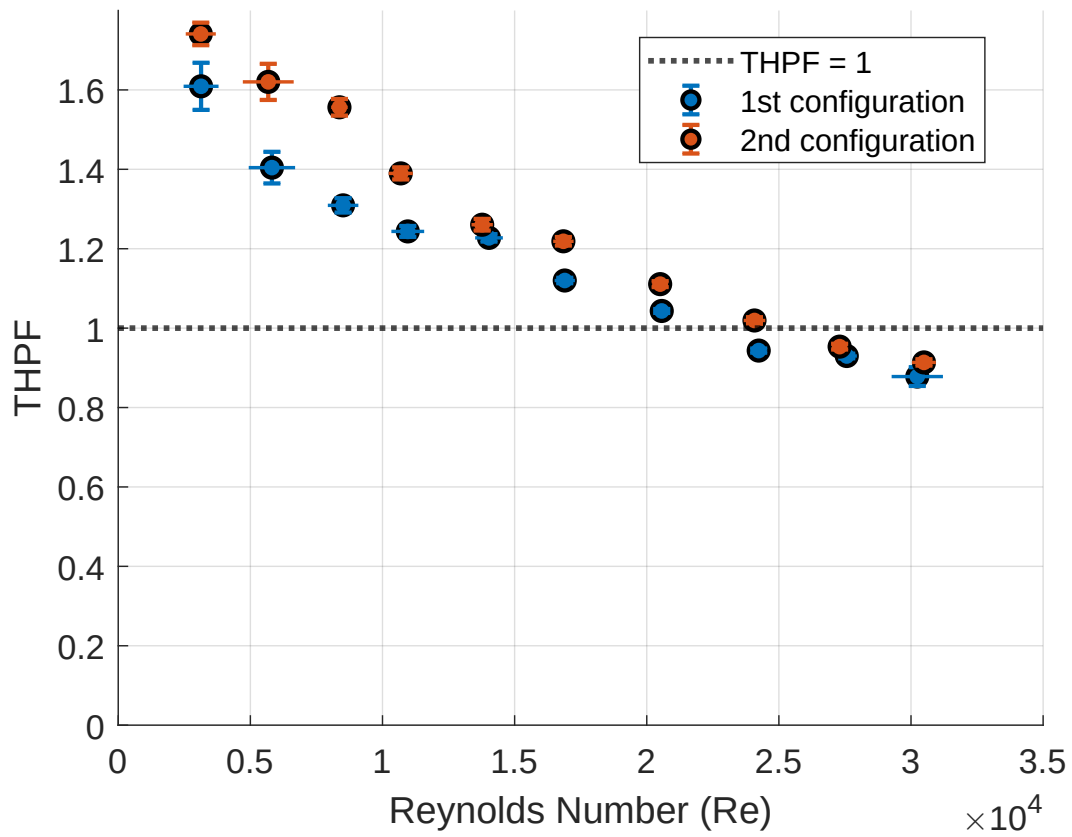


Fig. 12 Thermal-hydraulic performance factor (THPF) for the two configurations.

7. CONCLUSIONS

The present study experimentally investigated the thermo-hydraulic performance of solar air-heater channels enhanced with louvered-sheet configurations. After validating the measurement system using a smooth channel baseline, two louvered sheets positioned at one-third and two-thirds of the test-section height were evaluated to determine their influence on heat-transfer enhancement and flow resistance.

Results show that the introduction of louvered sheets significantly increases the Nusselt number compared to the smooth channel, particularly at low Reynolds numbers where the flow remains more sensitive to induced secondary mixing. The Nusselt-number ratio reached its maximum at the lowest tested Reynolds number, confirming the ability of the louvered sheets to effectively disrupt and re-energize the thermal boundary layer. However, this enhancement comes at the cost of increased pressure losses, as reflected by the elevated friction factor and friction-factor ratio, both of which rise with increasing Reynolds number.

When evaluating the combined thermal and hydraulic behavior using the thermal-hydraulic performance factor (THPF), the first configuration exhibited a peak THPF of approximately 1.6 at a Reynolds number of around 3000. Beyond a Reynolds number of roughly 21,000, the THPF fell below unity, indicating that the pressure losses introduced by the enhancement outweighed the corresponding heat-transfer benefits at higher flow rates.

For the second configuration, the maximum THPF reached 1.74 at a Reynolds number of about 3000, and the THPF dropped below unity only after approximately 24,000, suggesting a broader feasible operating range than that of the first configuration. Moreover, the second configuration consistently maintained a higher THPF across the entire tested Reynolds-number range. These results emphasize the importance of optimizing heat-transfer enhancement features to achieve a favorable balance between thermal improvement and hydraulic penalty.

Overall, the experimental findings confirm that louvered sheets can offer meaningful thermal performance improvements for solar air-heater systems, particularly under low to moderate Reynolds-number operation. These insights contribute to the development of optimized passive-flow configurations that are suitable for compact, energy-efficient thermal systems. Future work should explore geometric optimization of louver angles, spacing, and multi-layer arrangements, as well as the integration of additive-manufactured structures to further enhance performance while minimizing pressure penalties. In addition, Particle Image Velocimetry (PIV) will be employed to visualize the flow structures generated by the louvered sheets, providing deeper insight into vortex formation, boundary-layer disruption, and secondary-flow behavior. These experimental flow-field measurements will also serve as high-quality validation data for upcoming CFD simulations, enabling more accurate numerical modeling and guiding the development of optimized hybrid passive-enhancement designs.

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NOMENCLATURE

h	Heat transfer coefficient	(W/m ² ·K)	$\bar{T}_{out}$	Average outlet temperature	(K)
Q_{conv}	Convective heat transfer	(W)	D_h	Hydraulic diameter	(m)
A	Convective heat transfer area	(m ²)	k	Thermal conductivity	(W/m·K)
$\bar{T}_s$	Heated surface average temperature	(K)	Nu	Nusselt number	(–)
T_f	Average fluid temperature	(K)	V	Average flow velocity	(m/s)
$\dot{m}$	Mass flow rate	(kg/s)	L	Test-section length	(m)
C_p	Specific heat capacity	(J/kg·K)	H	Test-section height	(m)
$\bar{T}_{in}$	Average inlet temperature	(K)	W	Test-section width	(m)
			ΔP	Static pressure difference	(Pa)

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